

Active Learning for Learning-to-Defer

Human-Centric Artificial Intelligence - Project 3

AG News classification with simulated experts,
learning-to-defer, active learning, and additional Human-AI analysis

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1. Introduction

This project studies when an AI text classifier should keep its own prediction and when it should ask an expert. We use the AG News dataset with four classes: World, Sports, Business, and Sci/Tech. The work covers a baseline classifier, simulated experts, learning-to-defer, and active learning. We also implemented the optional human-expert interface and additional analyses to better understand Human-AI collaboration.

2. Baseline Classifier

Method and choice

We use TF-IDF features with Logistic Regression. TF-IDF represents each article using weighted word features, giving more importance to words that are frequent within an article but less common across the dataset. This is well suited to news topic classification because terms such as “tournament,” “market,” or “software” can provide strong signals for distinguishing between different news categories.

Logistic Regression was chosen for three reasons. First, it is a strong and fast baseline for sparse text features. Second, it is reproducible and easier to inspect than a large neural network. Third, it provides class probabilities. Those probabilities are essential later because classifier confidence, entropy, probability margin, confidence-based deferral, and entropy-based active learning all depend on them.

Results

The baseline reaches 92.37% test accuracy. Sports is the strongest class at 97.95% correctly classified, while Business is the most difficult at 89.05%. Because the classifier is already strong, expert involvement must be selective rather than automatic.

Class / model	Result
Overall TF-IDF + Logistic Regression	92.37% test accuracy
World	91.42% correctly classified
Sports	97.95% correctly classified
Business	89.05% correctly classified
Sci/Tech	91.05% correctly classified

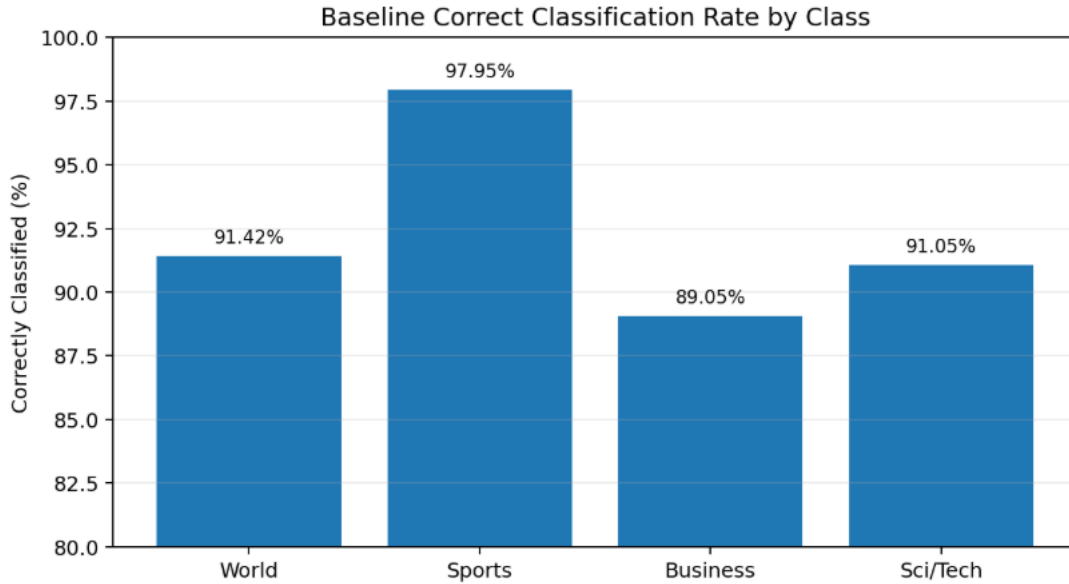


Figure 1. Correct Classification Rate by Class. The graph shows the percentage of test articles in each AG News class correctly classified by the baseline model.

3. Simulated Experts

Method and choice

The later experiments require repeated expert predictions, so we created two simulated experts. The experts are intentionally imperfect and have different levels of competence across the input space. A perfect expert would make the deferral problem trivial because it would almost always be preferable to ask the expert. Similarly, if an expert performed poorly on all types of articles, there would be little reason to ask for expert support. Therefore, we designed experts that are highly competent in specific regions but less reliable elsewhere.

The Sports and Business Specialist is strong on articles with clear Sports or Business vocabulary. The Technology and World Affairs Specialist is strong on Technology or World-affairs articles. At least two relevant keyword matches are required to enter a specialist region; otherwise the article belongs to the general region. When an expert is wrong, the incorrect class follows plausible confusion patterns rather than uniform random guessing.

Results and interpretation

Expert	Specialist regions	Observed specialist accuracy	General accuracy	Overall test accuracy
Sports/Business	Sports; Business	95.1%; 88.5%	56.9%	61.89%

Technology/World	Technology; World affairs	94.0%; 84.8%	56.3%	61.07%
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The results match the intended design. Both experts are very strong in their specialist regions but much weaker on general articles. Since most articles are general, overall expert accuracy is only about 61%. This creates the non-trivial collaboration problem; the expert can be useful but should not replace the AI globally.

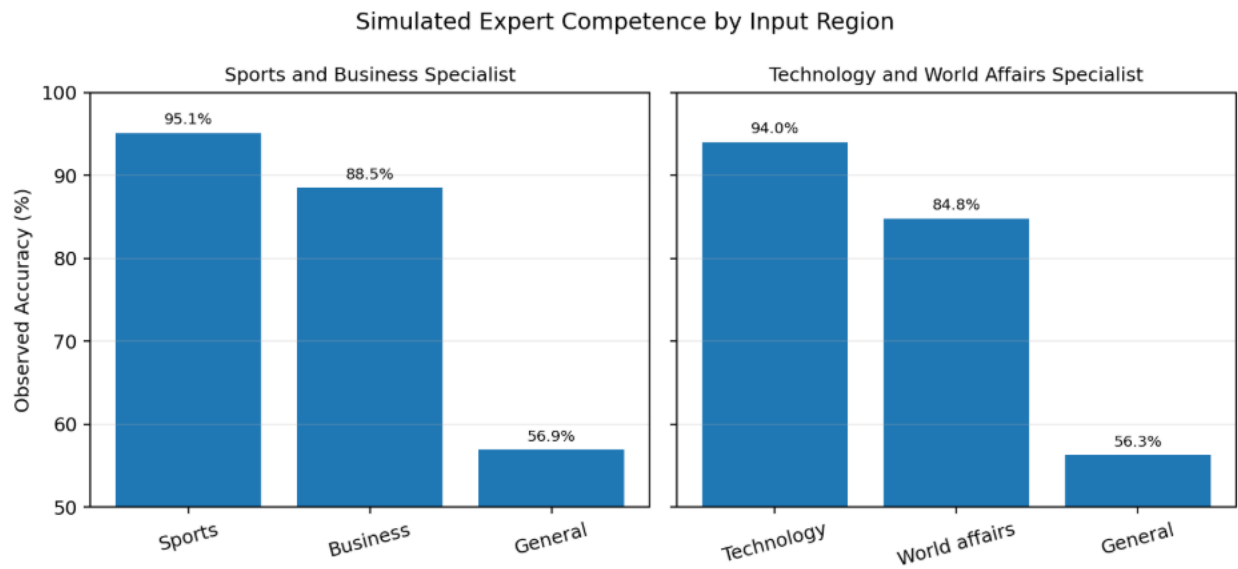


Figure 2. Simulated Expert Competence by Input Region. Both experts are strong in their specialist regions and substantially weaker on general articles.

4. Learning to Defer

Method and choice

For each article, the system chooses between the classifier prediction and the expert prediction. We compare two strategies. Confidence Threshold is the transparent baseline: it defers when classifier confidence is low. Competence-Aware is learned: it estimates when expert involvement is useful using classifier confidence, entropy, margin, text length, predicted class, expert region, and expected regional expert reliability.

Development data is separated into classifier-training, deferral-training, and validation parts (84,000 / 18,000 / 18,000 examples). Thresholds are selected on validation data rather than on the official test set. We evaluate both classification performance and deferral quality. A beneficial deferral is a case where the classifier is wrong and the expert is correct.

Results

Expert	Strategy	Team accuracy	Deferral rate	Beneficial precision	Missed beneficial
Sports/Business	Confidence Threshold	92.53%	3%	0.3122	298
Sports/Business	Competence-Aware	92.42%	3%	0.2798	306
Technology/World	Confidence Threshold	92.54%	1%	0.3208	314
Technology/World	Competence-Aware	92.63%	6%	0.2441	235

For the Sports/Business expert, Confidence Threshold performs best at 92.53%. For the Technology/World expert, Competence-Aware performs best at 92.63%. Therefore, the learned policy is not automatically superior; the best routing strategy depends on the expert's competence pattern.

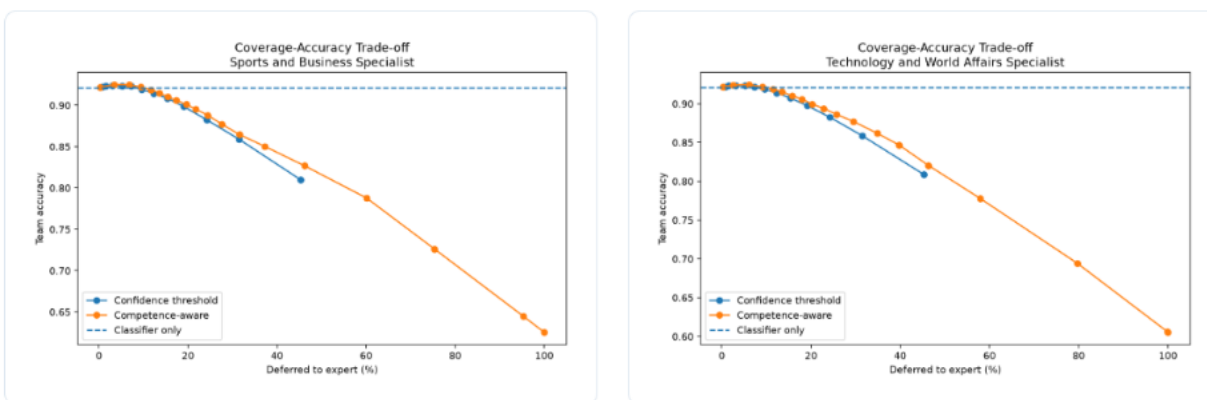


Figure 3. Coverage-Accuracy Trade-off for both simulated experts. The curves show validation-stage team accuracy as the deferral threshold is varied. The dashed line represents classifier-only performance.

The figure shows why the deferral rate is an important part of the evaluation. For both experts, the strongest performance occurs when only a small proportion of cases is deferred. As more articles are transferred to the expert, team accuracy eventually falls below the classifier-only baseline. This confirms that successful Human-AI collaboration depends on selective routing rather than frequent expert use.

Deferral quality adds information that accuracy alone cannot show. For example, the Technology/World competence-aware policy misses fewer beneficial opportunities (235) than the confidence baseline (314), but its beneficial-deferral precision is lower. It finds more opportunities to use the expert, while some extra deferrals are less precise.

5. Active Learning

Method and choice

Learning-to-defer assumes that expert responses are already available on development examples. Active learning makes the setting more realistic: initially there are no expert responses. The system has a limited query budget and must choose which articles to send to the expert. Each response becomes training data for a competence model.

We selected Classifier Entropy as our primary active learning strategy because it is suitable for the cold-start setting of this experiment. Initially, no expert responses are available, so the system cannot reliably estimate expert competence. In contrast, the classifier already provides a probability distribution for every unlabeled example. Entropy uses this information to prioritize examples on which the classifier is most uncertain. This makes the strategy simple, reproducible, and applicable from the first expert query without requiring prior knowledge of the expert.

We compare Entropy with Random Sampling, Competence Uncertainty, and Hybrid Uncertainty + Diversity. Random is an uninformed baseline. Competence Uncertainty queries cases where the current competence model is unsure. Hybrid adds diversity to reduce redundant queries. Results are recorded for budgets from 100 to 1,000 expert queries.

Results

Expert	Strategy	AUROC	F1	Team accuracy	Deferral
Sports/Business	Random	0.5382	0.5858	92.38%	0%
Sports/Business	Classifier Entropy	0.5500	0.7566	92.75%	4%
Sports/Business	Competence Uncertainty	0.5456	0.5968	92.24%	7%
Sports/Business	Hybrid	0.5567	0.6069	92.51%	4%
Technology/World	Random	0.5365	0.5678	92.39%	1%
Technology/World	Classifier Entropy	0.4671	0.6489	92.59%	3%
Technology/World	Competence Uncertainty	0.5018	0.5818	92.12%	9%

Technology/World	Hybrid	0.5382	0.5758	92.33%	5%
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At 1,000 queries, Classifier Entropy achieves the highest Human-AI team accuracy for both experts: 92.75% for the Sports/Business expert and 92.59% for the Technology/World expert. These results provide empirical support for our choice of Classifier Entropy as the primary active learning strategy. The comparison also shows that the strategy with the best competence-model metric does not necessarily produce the best final team accuracy. Therefore, we consider both competence-model performance and final Human-AI team performance when evaluating the active learning strategies.

The Classifier Entropy learning curve is not monotonic. For the Technology/World expert, team accuracy is 92.58% at 100 queries, 92.61% at 200, 92.53% at 400, 92.49% at 600, 92.53% at 800, and 92.59% at 1,000. This shows that increasing the number of expert queries does not automatically improve collaboration. The selected examples must also provide useful information for learning when expert involvement is beneficial.

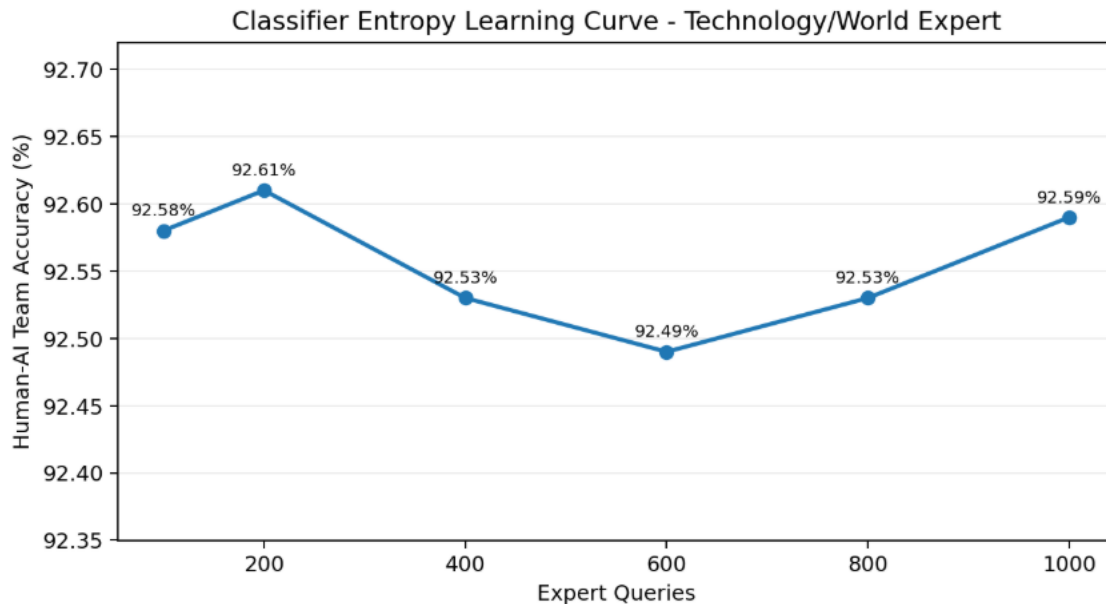


Figure 4. Classifier Entropy Learning Curve for the Technology and World Affairs Specialist. Team accuracy changes only slightly as the query budget grows and is not monotonic.

6. Additional Tasks and Advanced Analysis

We implemented additional analyses to explain why collaboration succeeds or fails rather than reporting only final accuracy.

Analysis	Why we added it	Main finding
Complementarity	Shows where AI and expert succeed on different examples.	Expert-only correct cases are 4.8% and 4.6% for the two experts.
Collaboration headroom	Compares the learned team with an ideal oracle.	Oracle accuracy is about 97%, so better routing is still possible.
Calibration	Checks whether confidence is reliable for deferral.	Expected Calibration Error is 0.0220.
Expert stability	Checks sensitivity to simulated-expert randomness.	Multiple random seeds are evaluated.
Active-learning efficiency	Compares strategies as query budget grows.	Performance depends on strategy, budget, and expert profile.
Ablation	Tests which information helps learned routing.	Feature groups are removed while the classifier is kept fixed.

Complementarity is especially important: only about 4.6-4.8% of test examples are cases where the expert is correct and the AI is wrong. These cases contain the main collaboration opportunity. The oracle accuracy of about 97% shows that the opportunity exists, but identifying the right cases before seeing the answer remains difficult.

Human-AI Complementarity

Overall expert accuracy is much lower than AI accuracy, but this does not mean the expert is useless. What matters for collaboration is whether the expert can solve some examples that the AI misses. We therefore divide the test set into four groups: both correct, AI only correct, expert only correct, and both wrong.

Expert profile	Both correct	AI only correct	Expert only correct	Both wrong
Sports/Business	57.3% (4,351)	35.1% (2,669)	4.8% (367)	2.8% (213)
Technology/World	56.6% (4,298)	35.8% (2,722)	4.6% (348)	3.1% (232)

The expert-only-correct cases are the most important collaboration opportunities. They account for 367 test articles for the Sports/Business expert and 348 for the Technology/World expert. A successful deferral policy should identify as many of these cases as possible without sending too many AI-correct cases to the expert.

Collaboration Headroom and Oracle

We also compare AI-only performance, expert-only performance, the best learned team, and an oracle. The oracle is not a deployable method: it assumes we already know whether the AI or expert will be correct for each article. Its purpose is to show the maximum benefit that perfect routing could obtain from the same AI and expert predictions.

Expert profile	AI alone	Expert alone	Best learned team	Oracle	Best deferral strategy
Sports/Business	92.37%	62.08%	92.53%	97.20%	Confidence Threshold
Technology/World	92.37%	61.13%	92.63%	96.95%	Competence-Aware

The oracle reaches about 97% accuracy for both experts, while the learned teams remain around 92.5-92.6%. This shows that useful complementary expert knowledge exists, but identifying the correct cases for deferral is the main difficulty. The additional analysis therefore supports the learning-to-defer result rather than simply reporting another accuracy number.

Confidence Calibration

Confidence is used by the Confidence Threshold strategy and classifier entropy is used by Active Learning, so we checked whether the baseline probabilities are reasonably calibrated. The Expected Calibration Error (ECE) is 0.0220. This low value indicates that classifier confidence is sufficiently reliable for confidence-based routing and uncertainty-based querying, although calibration does not guarantee that every uncertain case is a good case for expert deferral.

Expert Stability and Feature Ablation

The simulated experts contain stochastic behavior when producing incorrect predictions. To ensure that conclusions are not caused by one lucky random draw, the Advanced Analysis page evaluates expert behavior across repeated random seeds. We also implemented an ablation analysis for the competence-aware deferral model. Feature groups can be removed while the underlying text classifier remains fixed, allowing us to examine whether confidence information, uncertainty measures, predicted class, and expert-region information contribute to routing decisions.

These checks are supporting analyses rather than separate models. Their purpose is to test robustness and to make the learned deferral mechanism easier to understand.

Active-Learning Efficiency

The Active Learning experiment records intermediate results as the expert-query budget grows. This lets us evaluate sample efficiency rather than looking only at the final 1,000-query point. The learning curves show that performance is not monotonic: additional expert labels can change the

competence model and its deferral decisions without always increasing final team accuracy. For the Technology/World expert with Classifier Entropy, for example, team accuracy is 92.61% at 200 queries and 92.59% at 1,000 queries. This reinforces the Human-Centric AI point that the value of expert interaction depends on which examples are queried, not simply on the number of interactions.

7. Optional Human Expert Extension

We also implemented the optional interface where a real user acts as the expert. The user receives actively selected AG News articles, assigns one of the four labels, and responses are stored per session for later competence analysis.

The classifier prediction, confidence, entropy, ground-truth label, and correctness are hidden until the user has submitted a label. This reduces anchoring: showing the AI answer first could influence the human and make the collected response less independent.

8. Conclusion

The baseline classifier reaches 92.37% accuracy. The simulated experts are intentionally weaker overall but very strong in selected regions, creating a meaningful need for selective deferral. Learning-to-defer improves the team slightly over AI alone, but the best strategy depends on the expert: Confidence Threshold is best for Sports/Business, while Competence-Aware is best for Technology/World.

In the cold-start Active Learning setting, Classifier Entropy is practical because it can select queries before an expert model exists. At 1,000 queries it gives the highest team accuracy for both experts in our comparison. Overall, human involvement should be selective: the goal is not to use the expert as often as possible, but to identify cases where expert knowledge can actually improve the AI decision.